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Fault localization in active distribution networks: A novel power feature approach

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ABSTRACT

The increasing penetration of renewable energy sources in distribution networks introduces signal distortions and power quality issues that degrade the accuracy of phasor measurements used in monitoring and protection systems. In addition, dynamic loads and intermittent renewable generation can cause protection relay malfunctions, loss of coordination, and large-area outages. Consequently, fast faulty section localization and event identification are essential to enhance system reliability and enable rapid service restoration. To address these challenges, this paper proposes a novel set of power-based features extracted from sliding-window averages of instantaneous active and reactive power, along with the ratio of instantaneous active power to apparent power. These features remain stable under normal operating conditions but exhibit clear and localized variations during fault events, with increased sensitivity near the faulted bus. Such spatial characteristics enable accurate discrimination of faults between adjacent buses.A graph neural network (GNN)-based framework is developed to effectively integrate the extracted features with the network topology. The proposed approach is evaluated on a modified IEEE 33-bus active distribution network with photovoltaic integration under diverse operating scenarios. Validation using MATLAB/Simulink simulations, virtual real-time Typhoon HIL simulations, and field data confirms that the method outperforms conventional and existing learning-based techniques in fault localization accuracy and robustness.

1. Introduction

Accurate fault-section localization (FSL) in distribution networks (DNs) is essential for minimizing outage duration, reducing economic losses, enhancing service reliability, and improving customer satisfaction. Traditionally, after an outage, faulted sections are identified through manual feeder patrolling, which can take many hours to restore power. To expedite early restoration, measurement-based FSL was introduced in the early 2000s as smart grid technologies advanced [1–3]. With the deployment of communication infrastructure and intelligent electronic devices (IEDs), impedance-based methods and the identification of low-voltage zones became widely accepted [4]. While some limitations of conventional fault-location approaches have been addressed, the increased complexity of DNs with multiple laterals and topology changes to improve performance necessitates further advancements.

Furthermore, the increasing penetration of distributed energy generators (DEGs), particularly Inverter-based distributed generators (IBDGs), to reduce carbon emissions has introduced new challenges for FSL. The variation in fault current level depending on the location and type of DG connection has increased the risk of relay miscoordination. Additionally, various control schemes embedded in DEGs, which operate faster, have resulted in modulated fault voltage and current, leading to inaccurate FSL decisions even when advanced measurement systems are available [5,6].

Several improved FSL approaches have been reported in the literature. In [3,7], smart meter voltage data and load models are combined with impedance-based techniques to enhance localization accuracy. Nevertheless, the impact of high DEG penetration on these methods has not been thoroughly analyzed. A circuit-analysis-based FSL method using voltage-sag characteristics and both synchronized and unsynchronized measurements is proposed in [8]. While this approach improves FSL capability, voltage control techniques embedded in DEGs may result in small change in bus voltages which is similar to voltage changes during normal condition, making it difficult to locate the faulty section. A mixed-integer linear programming (MILP) based probabilistic method utilizing data from digital relays, IEDs, and smart meters is introduced in [9]. Although it provides probabilistic fault-location information, the computational burden can become significant for large-scale networks, and its robustness may be affected by rapidly changing operating conditions associated with DEG variability. Additionally, the

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Table 1
Comparative analysis of existing fault location and protection methods.

Ref.	System	Method	Benefits	Limitations
[4]	25-kV SaskPower feeder (Canada)	Impedance-based FL	Simple; low computation	low sensitivity to high R_f
[6]	Real utility test feeder	Short-circuit based FSL	Section narrowing + location	Requires accurate topology; weak DG validation
[8]	IEEE 7-bus system	Voltage-sag based FSL	No synchronized current needed	low sensitivity to high resistance fault
[9]	EPRI Ckt-7 feeder	Probabilistic data-driven FSL	Handles data/parameter uncertainty	Large dataset; high computation
[10]	IEEE 33-bus ADN	Phasor-feature based FSL	Suitable for DG systems	Limited inverter current-control analysis
[12]	6 bus microgrid	Converter control-aided protection	Enables conventional relays	Requires controller information
[15]	9 bus microgrid	Harmonic-based directional FSL	Effective under low fault current	Sensitive to background harmonics

method depends on operator interpretation for final decision-making and offers limited discussion on scalability under high DEG penetration. In [10], a fault-location strategy based on micro-phasor measurement units (μPMUs) and smart meters is presented, where the network is divided into zones to identify the faulted bus and line segment. However, with increased complexity of network topology and changes in network configuration may result in inaccurate FSL decisions.

High penetration of IBDGs has also led to increased harmonic distortion, particularly in triplen harmonics [11,12]. Prior studies have demonstrated that harmonic-domain features can provide valuable insights for fault analysis, even when fundamental components are severely distorted [13–15]. Nonetheless, existing smart-meter-based approaches continue to face challenges related to data quality, parameter uncertainty, and requent topological reconfigurations resulting from dynamic network operations.

As shown in Table I, the existing approaches have limited performance due to dynamic changes caused by DG penetration, sensitivity to topology changes, and dependent on accurate system parameters. To address these issues, recent studies have shifted toward AI-based FSL methods, which offer greater adaptability, robustness, and improved FSL accuracy in modern distribution networks. To overcome these limitations, several intelligent fault-localization methods have been proposed,including Gaussian process regression (GPR) and support vector machine (SVM) models [16], high-impedance fault detection using variational mode decomposition (VMD) [17], graph convolutional network (GCN)-based approaches [18], and hybrid time-frequency deep learning frameworks such as CWT-CNN and HHT-CNN [19,32]. However, HHT-based schemes often lack real-time feasibility, while CNN-based models tend to overfit when trained on limited datasets. Similarly, traditional artificial neural networks, one-dimensional CNNs [20], and DOST-MKELM-based methods [21] generally require large training datasets and high computational effort, and they do not explicitly model spatial correlations among neighbouring buses. Consequently, their performance degrades under noisy measurements and unseen operating conditions.

To address these challenges, this work proposes a new features for graph neural network (GNN) for FSL. By explicitly embedding network topology and pre-fault relationships into the learning process, the proposed approach effectively captures both local electrical measurements and the dependencies of the buses. This enables improved robustness to noise, stronger generalization across diverse operating conditions, and more reliable identification of faulted sections in ADNs with high IBDG penetration, elevated fault resistance, and significant triplen harmonic distortion. The main contributions of this work are summarized as follows:

1. This paper proposes a new set of power-based features for GNN-assisted fault localization, enabling accurate FSL in ADNs. The proposed features are sliding-window averages of the instantaneous active and reactive powers, as well as the ratio of the instantaneous active power to the apparent power.
2. A graph neural network (GNN) is employed to explicitly model the electrical topology of the distribution system and the spatial relationships among neighboring buses, resulting in improved fault localization accuracy and better interoperability compared with conventional machine learning approaches.
3. The proposed framework operates directly on time-domain measurements, thereby avoiding phasor estimation and reducing computational complexity, which makes the method suitable for real-time applications even with power quality degradation in the presence of IBDG.
4. The effectiveness and robustness of the proposed feature set are validated through offline simulations in MATLAB/Simulink 2023a and dataset generated by typhoon 2025.1.1 in virtual mode using the IEEE 33-bus distribution system. Additional validation is carried out using real field measurement data obtained from a distribution utility (DISCOM).
5. Performance is assessed under variations in unbalanced loading, load variations, high-resistance faults, harmonics, noise, IBDGs penetration, overloading, power factor variations, and diverse fault types.

The proposed methodology offers a scalable and robust solution for FSL in ADN with high penetration of IBDGs. The structure of this paper is organized as follows. Section 2 presents the detailed formulation of the proposed FSL methodology. Section 3 describes the performance evaluation procedures, followed by Section 4, which presents and discusses the obtained results. The proposed features and algorithm are validated through both offline and real-time studies using MATLAB/Simulink 2023a, Typhoon HIL 2025.1.1, and the Real field data. Finally, Section 5 concludes the paper and highlights potential avenues for future research.

2. Proposed faulty section localization methodology

2.1. Problem formulation, system description and data acquisition

Literature reports that the control schemes in IBDGs effectively mitigate variations in current and voltage magnitudes even under fault conditions. Therefore, time-domain features are advantageous than phasor-based features. With this in mind, this paper proposes an instantaneous power-based approach to improve FSL accuracy.

To illustrate the FSL problem, a modified IEEE 33-bus distribution network is modeled and simulated in the MATLAB/Simulink 2023a and Typhoon/editor window platform. The single-line diagram is shown in Fig. 1. The test system comprises 33 buses, 35 distribution lines, and 32 loads, along with three normally open (NO) switches to ensure radial operation of the originally meshed network. To emulate an active distribution scenario, PV-based distributed energy generators (PV-DEGs) are integrated at buses 18, 22, 25, and 33. The network is protected by a main feeder protection relay located at the Central Monitoring Unit (CMU) at the grid interconnection point (bus 1), while all downstream loads and PV-DEGs are equipped

with coordinated local protection relays to ensure selective and reliable fault isolation. The CMU relay performs supervisory decision-making, system-wide fault assessment, and restores the power follwing an outage, whereas the local relays operate as primary protective devices for rapid fault detection and isolation at the feeder and load levels.

Field Remote Terminal Units (FRTUs) and Intelligent Electronic Devices (IEDs) are deployed throughout the network with distinct functional roles. FRTUs primarily serve as communication and data acquisition interfaces, transmitting real-time measurements, breaker status,

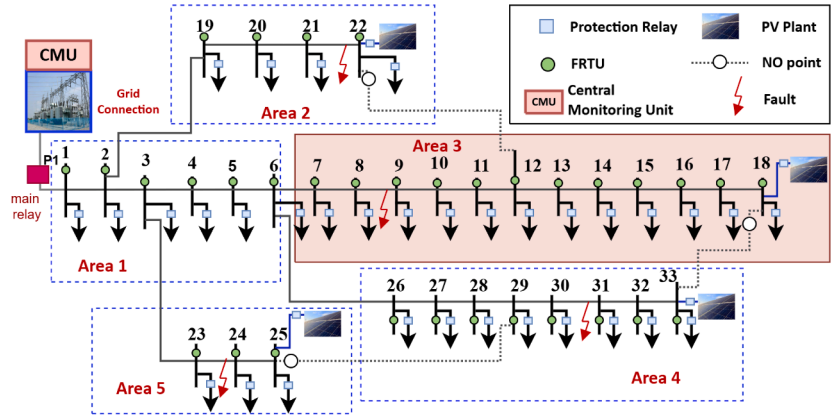


Fig. 1. Enhanced single line diagram of IEEE 33 bus system.

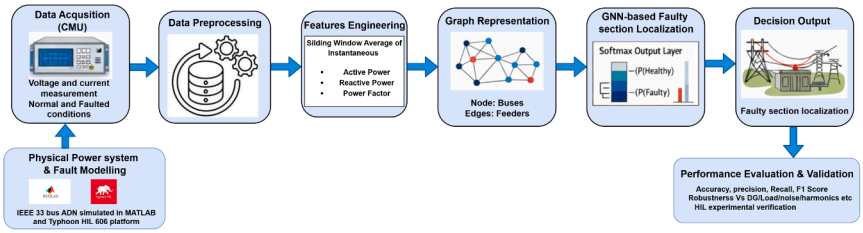


Fig. 2. Flow diagram of proposed methodology.

and switch positions to the CMU. In contrast, IEDs perform local protection, control, and relay logic functions while also recording high-resolution electrical measurements. Data exchange between field devices and the Advanced Distribution Management System (ADMS) is enabled through standardized communication protocols such as DNP3 over TCP/IP and IEC 60870-5-101/104, ensuring reliable, interoperable, and low-latency operation. Modern IEDs support both steady-state monitoring and transient waveform recording with sampling rates up to 1 kHz, enabling precise capture of fault signatures and system disturbances [22]. During fault events, instantaneous line current and voltage waveforms recorded by the IEDs are transmitted to the CMU, forming a large-scale dataset for feature extraction and machine learning-based classification. These discriminative features enable accurate and fast faulted section localization in ADNs under varying operating conditions.

The flow diagram of proposed methodology is illustrated in Fig. 2, here three-phase voltages and currents datasets are generated under normal and faulted operating conditions using the IEEE 33-bus test feeder modeled in MATLAB/Simulink 2023a and Typhoon/editor window (model HIL 606) in virtual mode. The raw signals are preprocessed using sliding-window averaging to suppress noise and transient distortions, from which three-phase instantaneous active power, reactive power, and power factor are computed and structured as node feature vectors, with buses represented as graph nodes and distribution lines as edges to preserve network topology. These graph-embedded features are processed by a GNN through neighborhood message passing, enabling joint learning of electrical connectivity and spatial correlations associated with fault propagation. The trained network outputs a probability distribution over candidate feeder sections via a Softmax layer, facilitating accurate localization of the faulted segment. Detailed descriptions of the feature-engineering pipeline and the adopted GNN architecture are provided in the corresponding subsections, while comprehensive validation across multiple platforms (MATLAB, Typhoon 606, Real field dataset), and operating scenarios with parameter variations is presented in the Results section.

2.2. Feature engineering

In ADN, continuous variations in operating conditions, including load dynamics, measurement noise, and distributed generation intermittency, significantly degrade the performance of conventional fault localization methods that rely on system parameter and capacity assumptions. To overcome the aforementioned limitations, this study introduces a set of robust and discriminative instantaneous power-based features extracted using a sliding-window averaging strategy. The proposed features facilitate reliable FSL in ADN under diverse and time-varying operating conditions.

A sliding-window averaging method is utilized to mitigate the impact of measurement noise, short-term variances and temporal disturbances and maintain fault contributing properties. The resulting features have better stability and reliability, which are useful in proper fault detection and faulted section localization. To localize a fault section, monitored buses and feeders provide three phase voltage and current signals in the time domain, which are used to localize a fault section. The signals are handled to calculate instantaneous power features that describe the behavior of the system in both normal and faulted operating conditions. These signals are processed to compute instantaneous power features that characterize system behavior under normal and faulted operating conditions. Consider a three-phase distribution line ℓ with phase voltages $v^{\phi}(n)$ and currents $i_{\ell}^{\phi}(n)$, where $\phi \in \{R, Y, B\}$ denotes the phase and n represents the discrete-time index. The instantaneous active and reactive power of each phase is determined as

$$P_{\ell}^{\phi}(n) = v^{\phi}(n)\, i_{\ell}^{\phi}(n), \tag{1}$$

$$Q_{\ell}^{\phi}(n) = v^{\phi}(n)\, \mathcal{H}\left\{ i_{\ell}^{\phi}(n) \right\}, \tag{2}$$

where $\mathcal{H}\{\cdot\}$ denotes the Hilbert transform, which generates a 90° phase-shifted version of the current signal. This phase shift allows for the instantaneous separation of active and reactive power components in time-domain analysis.

The total instantaneous three-phase active and reactive powers are obtained by summing over all phases:

$$P_{\ell}^{\text{tot}}(n) = \sum_{\phi\in\{R,Y,B\}} P_{\ell}^{\phi}(n), \tag{3}$$

$$Q_{\ell}^{\text{tot}}(n) = \sum_{\phi\in\{R,Y,B\}} Q_{\ell}^{\phi}(n). \tag{4}$$

The instantaneous power factor of the line is defined as the ratio of total active power to apparent power:

$$\text{PF}_{\ell}(n) = \frac{P_{\ell}^{\text{tot}}(n)}{\sqrt{\left(P_{\ell}^{\text{tot}}(n)\right)^2 + \left(Q_{\ell}^{\text{tot}}(n)\right)^2}}. \tag{5}$$

To suppress high-frequency fluctuations while preserving essential dynamics, a sliding-window average of width W samples is applied:

$$P^{\text{SW}}(n) = \frac{1}{W}\sum_{k=n-W+1}^{n} P^{\text{tot}}(k), \tag{6}$$

$$Q^{\text{SW}}(n) = \frac{1}{W}\sum_{k=n-W+1}^{n} Q^{\text{tot}}(k), \tag{7}$$

$$\text{PF}^{\text{SW}}(n) = \frac{1}{W}\sum_{k=n-W+1}^{n} \text{PF}(k), \tag{8}$$

where $P^{\text{tot}}(k), Q^{\text{tot}}(k)$, and $\text{PF}(k)$ represent the instantaneous active power, reactive power, and power factor at the kth discrete-time sample, respectively.The parameter W denotes the length of the sliding-window that determines the amount of temporal smoothing of the signals. A window size of $W = 20$ samples or a duration is chosen in this work, 20 ms at a sampling rate of 1 kHz. A Hilbert-transform-based method is employed to compute reactive power by deriving the quadrature current component without relying on phasors [23]. A sliding-window average yields a constant active power under normal conditions, which decreases during a fault. The oscillatory instantaneous reactive power averages to zero over one cycle under steady-state operation but increases during a fault. These variations in active and reactive power are reflected in the instantaneous power factor, creating a distinct signature of disturbance with significant deviations following fault conditions. The impact of spurious signals are attenuated by summing the quantities of power across a moving sliding window. This improves robustness with measurement noise, data uncertainty, and transient operating conditions. improving fault section localization in live distribution networks. The derived inputs fed to the graph neural network framework are then in form of features. The following section outlines the structure and training procedure.

2.3. Graph neural network-based faulted section localization

A graph-based learning approach is employed for faulted section identification by incorporating both electrical measurements and network topology. The active distribution network is modeled as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ represents the set of buses and $\mathcal{E}$ denotes the feeders interconnecting the buses. Each bus $v \in \mathcal{V}$ is assigned a feature vector formed from sliding-window averaged electrical quantities, providing a compact description of the local operating state. For node v at discrete time index n, the input feature vector is defined as

$$\mathbf{x}_v(n) = \begin{bmatrix} P_v^{\text{SW}}(n) \\ Q_v^{\text{SW}}(n) \\ \text{PF}_v^{\text{SW}}(n) \end{bmatrix}, \tag{9}$$

where $P_v^{\text{SW}}(n), Q_v^{\text{SW}}(n)$, and $\text{PF}_v^{\text{SW}}(n)$ denote the sliding-window averaged active power, reactive power, and power factor at bus v, respectively.

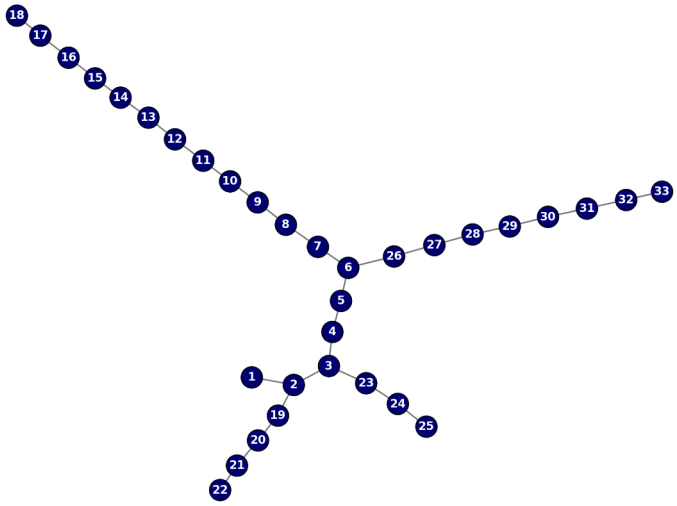


Fig. 3. Graph representation of the IEEE 33-bus distribution system.

These features offer a noise-resilient and temporally consistent description of system behavior under both normal and faulted operating conditions. The GNN updates node embeddings through an iterative message-passing process that aggregates information from neighboring nodes. At the lth layer, the embedding update for node v is expressed as

$$\mathbf{h}_v^{(l+1)} = \sigma\left(\sum_{u\in\mathcal{N}(v)} \mathbf{W}^{(l)}\mathbf{h}_u^{(l)} + \mathbf{b}^{(l)}\right), \tag{10}$$

where $\mathbf{h}_v^{(l)}$ denotes the embedding of node v at layer l (with $\mathbf{h}_v^{(0)} = \mathbf{x}_v$), $\mathcal{N}(v)$ represents the set of neighboring nodes, $\mathbf{W}^{(l)}$ and $\mathbf{b}^{(l)}$ are trainable parameters, and $\sigma(\cdot)$ is a nonlinear activation function.Through successive neighborhood aggregation, the GNN captures both localized electrical disturbances and their spatial propagation across the network. The final-layer embeddings are mapped to class probabilities using a softmax classifier, given by

$$\hat{\mathbf{y}}_v = \text{Softmax}\left(\mathbf{W}_o\mathbf{h}_v^{(L)} + \mathbf{b}_o\right), \tag{11}$$

where $\mathbf{h}_v^{(L)}$ denotes the output embedding of the last GNN layer and $\hat{\mathbf{y}}_v$ represents the predicted fault class for node v.By integrating the temporally smoothed power features with the network topology, the proposed GNN framework enables accurate and robust faulted section localization under a wide range of operating conditions.

2.4. Graph-based network representation

The IEEE 33-bus radial distribution system is modeled as an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ denotes the set of buses ($|\mathcal{V}| = 33$) and $\mathcal{E}$ the set of connecting line sections ($|\mathcal{E}| = 32$), as illustrated in Fig. 3. The topology is encoded by a symmetric binary adjacency matrix

where $A_{ij} = A_{ji}$ and $A_{ii} = 0$. Each node is assigned a feature vector $\mathbf{x}_i = [P_i^{\text{SW}}, Q_i^{\text{SW}}, \text{PF}_i^{\text{SW}}]^\top \in \mathbb{R}^3$, comprising sliding-window averaged active power, reactive power, and power factor. The resulting graph input ($\mathbf{A}$, $\mathbf{X} \in \mathbb{R}^{33\times3}$) jointly encodes network topology and local electrical operating conditions for faulted section localization.

2.5. Proposed GNN model architecture

The architectural details of the proposed GNN model are presented in Table 2 as per [33]. The model consists of three GNN layers with progressively increasing feature dimensions ($3 \to 16 \to 32 \to 64$), followed by global mean pooling to aggregate node-level embeddings into a graph-level representation. Two fully connected layers then map this representation to the final classification output.

Table 2
Proposed GNN model architectural details.

Layer no.	Layer detail	Feature size	Output shape	Trainable parameters
L_0	Input node features (P^{SW}, Q^{SW}, PF^{SW})	1×3	33×3	–
L_1	GNN layer + ReLU + Batch Normalization	$3 \rightarrow 16$	33×16	128
L_2	GNN layer + ReLU + Batch Normalization	$16 \rightarrow 32$	33×32	1088
L_3	Dropout ($p = .3$)	–	33×32	–
L_4	GNN layer + ReLU activation	$32 \rightarrow 64$	33×64	4160
L_5	Global Mean Pooling (graph-level embedding)	–	1×64	–
L_6	Fully Connected + ReLU	$64 \rightarrow 32$	1×32	2080
L_7	Fully Connected + Softmax (3 classes)	$32 \rightarrow 3$	1×3	99

GNN – Graph Neural Network, ReLU – Rectified Linear Unit, and Batch Normalization stabilizes training. Global Mean Pooling aggregates node-level embeddings into a graph-level representation.

2.6. Training and optimization

The model is trained in a supervised manner using labeled datasets containing fault events (FE), non-fault events (NFE), and healthy events (HE) generated under various system operating conditions. The training objective minimizes the categorical cross-entropy loss between predicted outputs and ground-truth class labels. The Adam optimizer is employed for parameter optimization, ensuring stable and efficient convergence. Regularization techniques, including dropout ($p = .3$) and early stopping, are applied to prevent overfitting and enhance generalization performance.

2.7. Computational complexity and inference time

The proposed GNN model comprises a total of 7555 trainable parameters with an approximate computational cost of 85,600 floating-point operations (FLOPs ≈ 0.086 MFLOPs) per inference. This is derived from the message-passing complexity of $\mathcal{O}(E \times F_{\text{in}} \times F_{\text{out}})$ per GNN layer, where $E = 32$ edges for the 33-node network. The layer-wise breakdown is: $L_1 = 32 \times 3 \times 16 = 1{,}536$ FLOPs, $L_2 = 32 \times 16 \times 32 = 16{,}384$ FLOPs, and $L_4 = 32 \times 32 \times 64 = 65{,}536$ FLOPs for the GNN layers. The fully connected layers contribute $L_6 = 64 \times 32 = 2{,}048$ FLOPs and $L_7 = 32 \times 3 = 96$ FLOPs, yielding a total of approximately 85,600 FLOPs.

On a standard industrial PC, the estimated inference time is 5–20 milliseconds per prediction. This holds critical practical importance: conventional fault location by field crew patrol typically requires 4 to 24 h of manual inspection across the feeder, during which customers remain de-energized and field personnel are exposed to occupational hazards. The proposed model therefore reduces fault location time by over 99.99%, as expressed by:

$$\Delta T = T_{\text{manual}} - T_{\text{GNN}} \approx (4 \text{ to } 24 \text{ hours}) - (< 20 \text{ ms}) \approx 99.99\% \text{ reduction} \tag{12}$$

This enables control room operators to dispatch field crews directly to the identified faulted zone, significantly reducing outage duration and improving the System Average Interruption Duration Index (SAIDI).

3. Performance evaluation metrics

The performance of the proposed GNN-based fault classification framework is evaluated using a confusion matrix (CM), which quantifies its ability to distinguish among FE, NFE, and HE classes in an active distribution network. For an n-class classification problem, the CM is defined as

$$\mathbf{CM} = \begin{pmatrix} CM_{11} & \cdots & CM_{1n} \\ \vdots & CM_{ij} & \vdots \\ CM_{n1} & \cdots & CM_{nn} \end{pmatrix}, \tag{13}$$

where CM_{ij} denotes the number of samples belonging to class i that are predicted as class j, and the diagonal elements CM_{ii} represent correctly classified instances.Based on the CM, overall classification accuracy is computed as

$$\text{Accuracy} = \frac{\sum_c (TP_c + TN_c)}{\sum_c (TP_c + TN_c + FP_c + FN_c)}. \tag{14}$$

To assess global classification performance, micro-averaged precision, recall, and F1-score are defined as

$$\text{Precision}_\mu = \frac{\sum_c TP_c}{\sum_c (TP_c + FP_c)}, \tag{15}$$

$$\text{Recall}_\mu = \frac{\sum_c TP_c}{\sum_c (TP_c + FN_c)}, \tag{16}$$

$$\text{F1}_\mu = 2 \times \frac{\text{Precision}_\mu \times \text{Recall}_\mu}{\text{Precision}_\mu + \text{Recall}_\mu}. \tag{17}$$

To address class imbalance, weighted-averaged metrics are employed and expressed as

$$\text{Precision}_w = \sum_c w_c \, \text{Precision}_c, \quad \text{Recall}_w = \sum_c w_c \, \text{Recall}_c, \quad \text{F1}_w = \sum_c w_c \, \text{F1}_c, \tag{18}$$

where the class weight w_c is defined as

$$w_c = \frac{N_c}{\sum_c N_c}, \tag{19}$$

and N_c denotes the number of samples in class c. Collectively, these metrics provide a comprehensive and reliable evaluation of the proposed GNN-based faulted section localization framework under diverse operating conditions.

4. Results and discussions

This section presents the performance evaluation of the features described in Section 2 and discusses the results obtained using the proposed FSL algorithm. The validation is carried out through both offline and online simulation studies under diverse operating scenarios.

4.1. Features performance validation

4.1.1. Using offline simulation

The performance of the proposed features is first validated using offline simulations conducted on the modified IEEE 33-bus distribution system in MATLAB/Simulink 2023a platform, as illustrated in Fig. 1. The system has a total rated active and reactive power demand of 3.715 MW and 2.3 MVAr, respectively. PV based DEGs are integrated at buses 18, 22, 25, and 33 with capacities of 350 kW, 400 kW, 350 kW, and 300 kW, respectively, resulting in a total installed capacity of 1.4 MW, which corresponds to a penetration level (PL) of 37.68% [24].

To assess the discriminatory capability of the extracted features, a double line to ground (LLG) fault, is simulated on line 8–9 in the vicinity of bus 8. Transient current signals recorded from all FRTUs are processed

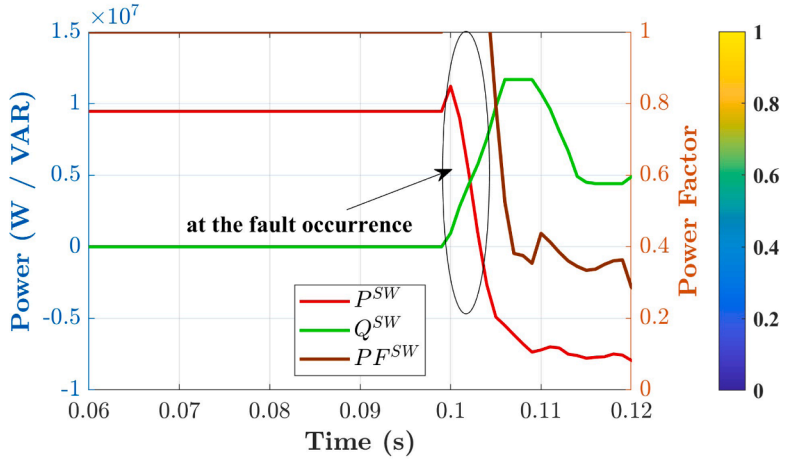


Fig. 4. Proposed feature of bus 8.

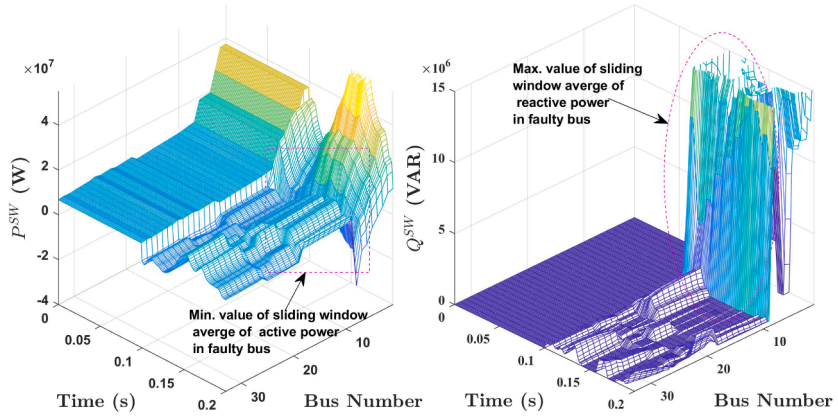


Fig. 5. Proposed features of 33 buses.

to extract the features $\mathcal{F}_1, \mathcal{F}_2$, and $\mathcal{F}_3$. Fig. 4 illustrates the variation of feature $\mathcal{F}_1, \mathcal{F}_2$, and $\mathcal{F}_3$ for a LLG fault , as observed at the near bus 8 of the network.

Fig. 4 illustrates the P^{SW}, Q^{SW}, and $\mathrm{PF}^{\mathrm{SW}}$ at Bus 8 of the IEEE 33-bus distribution system, simulated in the MATLAB platform, where a fault is applied on the line segment between Bus 8 and Bus 9. Prior to the fault inception at $t = 0.1$ s, the system operates under steady state conditions, characterized by a nearly constant P^{SW}, zero Q^{SW}, indicating balanced and healthy network operation. At the fault instant, a pronounced disturbance is observed, where the P^{SW} exhibits a sharp decline, the Q^{SW} increases significantly due to fault-induced reactive demand, and the power factor drops rapidly. These abrupt deviations in the sliding-window averaged features clearly capture the transient fault dynamics. The magnitude and temporal evolution of these variations provide highly discriminative signatures for FSL.

The P^{SW} and Q^{SW}, measured at all 33 buses are analyzed to assess system behavior under fault conditions, as illustrated in Fig. 5. A fault is initiated at $t = 0.1$ s on the line section between Bus 8 and Bus 9, resulting in noticeable variations in power responses across the network. Prior to fault occurrence, both P^{SW} and Q^{SW} remain approximately constant, indicating steady-state operation. After fault initiation, significant deviations appear in the active and reactive power profiles at several buses, reflecting the spread of fault-related disturbances through the system.

As shown in Fig. 5, the Q^{SW} reaches higher values at the upstream buses, from Bus 1 to Bus 8, while a clear decrease is observed beyond the faulted section, that is, from Bus 9 onward. This behavior is associated with increased reactive power demand and voltage support requirements near the fault location. In comparison, the P^{SW} exhibits a pronounced minimum at the faulted bus at the time of fault initiation, indicating a disruption in power transfer along the feeder. The resulting spatial variation of instantaneous active and reactive power along the feeder provides a clear signature of the fault location.

4.1.2. *Using online simulation in virtual mode*

The structural layout of the online simulation in the editor window of Typhoon HIL Control Center using virtual mode is illustrated in Fig. 6. The simulation is carried out using the same system parameters and operating conditions discussed in Result Section 4.1.1. The modified IEEE 33-bus distribution system is implemented in the Typhoon 606 editor environment and executed under real-time conditions. During simulation, the three-phase voltage and line current signals are continuously recorded using the embedded data streaming logger.

The Typhoon 606 is connected to an external workstation through a TCP/IP communication link for continuous data transfer and storage. The measured three-phase voltages (V_a, V_b, V_c) and currents (I_a, I_b, I_c) are imported into MATLAB through the TCP/IP protocol. In the MATLAB environment, the features $P^{\mathrm{SW}}, Q^{\mathrm{SW}}$, and PF^{SW} are computated and stored in CSV file for further processing in Jupyter Notebook. The generated dataset is then used for FSL using graph neural network (GNN) classification. Finally, the performance of the proposed model is evaluated using confusion matrix indices based on (13)–(18).

Fig. 7 illustrates the electrical response and extracted features at Bus 8 under faulted conditions. Subfigures (a) and (b) show the three-phase bus voltages (V_a, V_b, V_c) and currents (I_a, I_b, I_c), respectively. Before fault inception at $t = 0.1$ s, the waveforms are balanced and sinusoidal, indicating normal operating conditions. After the fault, pro-

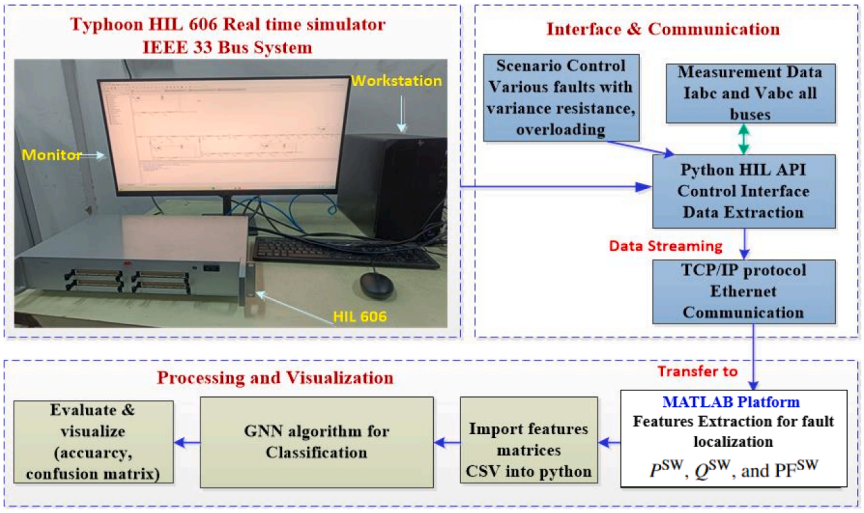


Fig. 6. Flow diagram of methodology in Typhoon HIL platform.

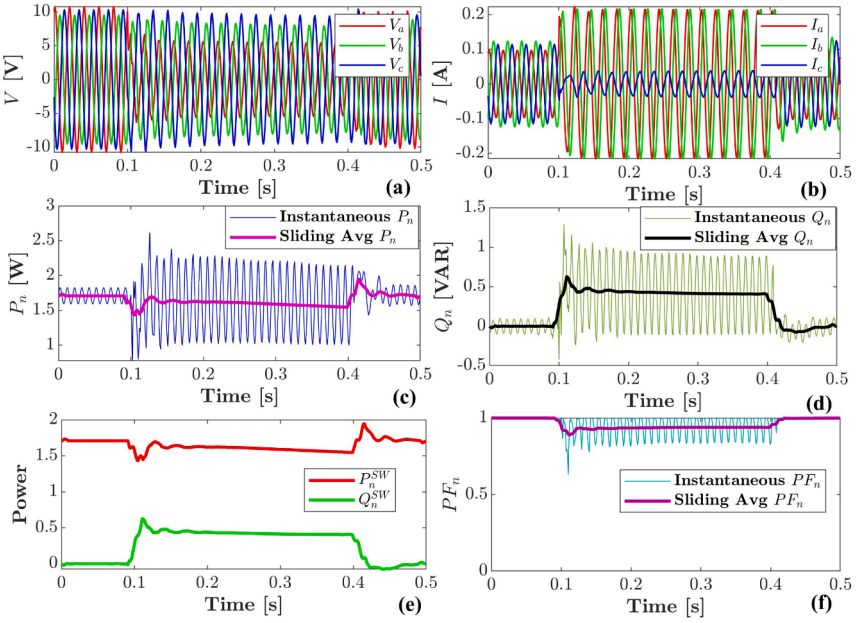


Fig. 7. Proposed feature of bus 8, using typhoon data sets.

nounced distortions and magnitude variations appear, particularly in the current signals, reflecting altered power flow behavior at Bus 8.

Subfigure (c) presents the instantaneous active power P_n and its sliding-window averaged counterpart P_n^{SW}. While P_n contains significant high-frequency oscillations, the sliding-window averaging effectively attenuates these components and highlights the dominant trend. A sharp decline in P_n^{SW} is observed immediately following the fault, indicating a disruption in real power transfer at the faulted bus. This characteristic variation provides a reliable indicator for fault proximity and localization.

Similarly, subfigure (d) depicts the instantaneous reactive power Q_n and its sliding-window averaged value Q_n^{SW}. Under pre-fault conditions, the reactive power remains close to zero, consistent with near-unity power factor operation. Following fault inception, a pronounced increase in Q_n^{SW} is observed, indicating elevated reactive power demand caused by the disturbance. Subfigure (e) compares the sliding-window averaged active and reactive powers, P_n^{SW} and Q_n^{SW}, respectively. Their opposing trends during the fault interval are clearly evident, with P_n^{SW} decreasing and Q_n^{SW} increasing, thereby providing complementary information for fault characterization.

Finally, subfigure (f) shows the instantaneous power factor PF_n and its sliding-window averaged value PF_n^{SW}. Prior to the fault, PF_n^{SW} remains close to unity, whereas a noticeable reduction occurs after fault inception due to the combined effect of reduced active power and increased reactive power. Overall, the sliding-window averaged features P_n^{SW}, Q_n^{SW}, and PF_n^{SW} at Bus 8 serve as stable and discriminative indicators for capturing the temporal characteristics of the fault.

The Fig. 8 presents the temporal variation of active power, reactive power, and power factor at Bus 8 and along Lines 9–8 and 8–9 under faulted operating conditions. The sliding window averaged responses are superimposed on the instantaneous signals to highlight the underlying trends. A fault is initiated at $t = 0.1$ s, resulting in distinct deviations in power flow characteristics.

At Bus 8, the sliding window averaged active power P_8^{SW} exhibits a noticeable reduction following fault inception, while the corresponding reactive power Q_8^{SW} increases due to elevated reactive power demand

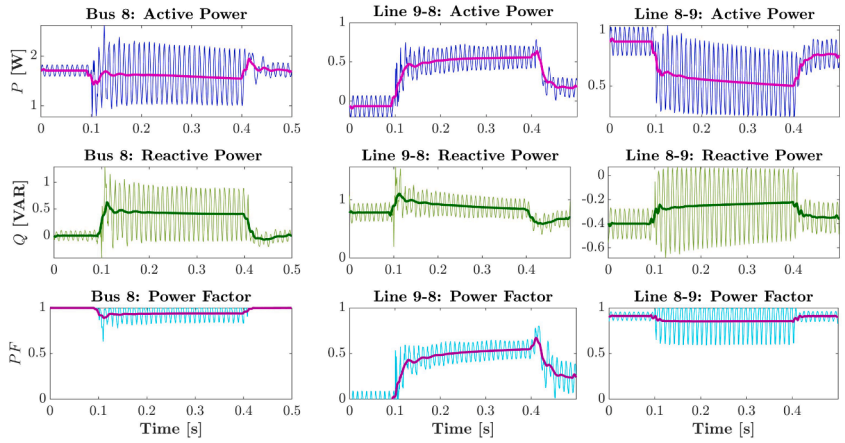


Fig. 8. Features of bus 8 and line connected to bus 8.

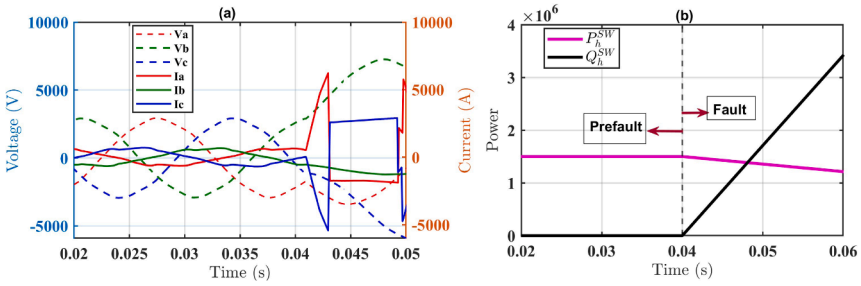


Fig. 9. Feature validation using real dataset.

near the fault location. Consequently, the power factor PF_8^{SW} deviates from unity during the faulted interval. Along line 9–8, both P^{SW} and Q^{SW} show increased magnitudes, indicating upstream power contribution toward the fault. In contrast, Line 8–9 exhibits reduced active power and altered reactive power flow, reflecting the downstream impact of the faulted section.The contrasting behaviors of the sliding window averaged active power, reactive power, and power factor across Bus 8 and the adjacent lines provide clear signatures of the fault.

4.1.3. Features validation using real field datasets

The time domain behavior of three-phase voltages and currents, along with the corresponding sliding-window averaged power features, is analyzed using a real field dataset provided by ODISHA DISCOM for a medium-voltage distribution feeder operating at a nominal frequency of 50 Hz. The measurements are recorded over a time duration of approximately 0.02–0.05 s, covering nearly 2.5 electrical cycles. Under healthy operating (0.02- 0.04 s) conditions, as shown in Fig. 9(a), the three-phase voltages (V_a, V_b, V_c) exhibit balanced sinusoidal waveforms with peak magnitudes of about ±2 kV, while the corresponding line currents (I_a, I_b, I_c) remain balanced with peak values around ±2 kA, indicating stable feeder loading and near-unity power factor operation. During a double-line-to-ground (LLG) fault at the instant of 0.04 s, severe voltage sag, waveform distortion, and abrupt current escalation are observed in the faulted phases, with current magnitudes reaching the order of 10^4 A, whereas the non-faulted phase retains partial voltage support, which is characteristic of LLG fault conditions in distribution systems. The sliding-window averaged active and reactive power features shown in Fig. 9(b) further emphasize this transition, where the P^{SW} remains nearly constant and the Q^{SW} stays close to zero during normal operation, while following fault inception a pronounced reduction in P^{SW} and a significant increase in Q^{SW} are observed due to voltage collapse and the dominance of reactive fault currents. These results demonstrate the suitability of sliding-window power-based features derived from real field measurements for reliable discrimination between NE and FE operating states in practical distribution networks.

4.2. Model performance indics

This section evaluates the performance of the proposed method using standard CM based performance indices defined in Section 2, under a wide range of operating conditions to assess robustness and generalization capability. Comprehensive fault and non-fault scenarios are generated in the MATLAB/Simulink platform by systematically varying fault types, fault locations, fault resistances, and DG penetration, as summarized in Table 3. Specifically, fault resistance is varied from 0.1 Ω to 100 Ω while considering ten distinct fault types, including single line to ground (SLG), line to line (LL), double line to ground(LLG), and three phase (LLLG) faults, at multiple locations along the distribution feeder. Furthermore, system loading conditions are diversified by varying load magnitude, power factor, and phase unbalance, and the impact of DEGs is incorporated by altering their penetration levels across multiple operating points. This extensive scenario generation ensures sufficient dataset diversity for reliable training and rigorous performance evaluation of the proposed FSL framework.

Based on the operating scenarios summarized in Table 3, diverse datasets were generated to compute the proposed instantaneous power–based features, which were subsequently provided to the GNN classifier.The dataset is generated using the IEEE 33-bus distribution system, where each bus is represented by three features, namely P^{SW}, Q^{SW}, and PF^{SW}. These features are obtained using a sliding window averaging (SWA) technique with a window size of 20 samples at a sampling frequency of 1 kHz, corresponding to 20 ms of data. The sliding window is shifted by one sample, forming successive windows (1–20, 2–21, …, 10–29). Each window represents a short-term temporal behavior of the system, and the effective shift between successive windows is 1 ms, while the relative shift between the first and tenth window is 9 ms. Thus, the total dataset size is computed as $33 \times 3 \times \text{Scenarios} \times 10$, where 33

Table 3
Description of generated operating and fault scenarios for the IEEE 33-bus distribution system.

Parameters	Details	Scn.	Generated Dataset Size
Fault resistance	0.1, 0.5, 1, 2, 5, 10, 20, 50 and 100 Ω	9	$33 \times 3 \times 9 \times 10 = 8910$
Fault type	ag, bg, cg, ab, bc, ca, abg, bcg, acg and abcg	10	$33 \times 3 \times 10 \times 10 = 9900$
Fault position	8–9, 21–22, 23–24 and 30–31	4	$33 \times 3 \times 4 \times 10 = 3960$
Load variation	(i) LC varied from 80% to 120% in steps of 10%	5	$33 \times 3 \times 5 \times 10 = 4950$
	(ii) LPF varied from 0.85 to 1.0	4	$33 \times 3 \times 4 \times 10 = 3960$
	(iii) UC loading with ±3%, ±6% and ±12%	6	$33 \times 3 \times 6 \times 10 = 5940$
DGs penetration (DGP) level (%)	0, 20, 30, 35, 40 and 45	6	$33 \times 3 \times 6 \times 10 = 5940$

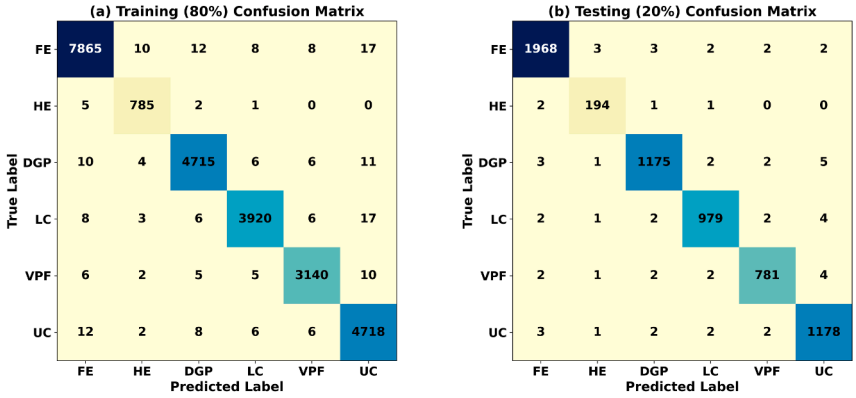


Fig. 10. Confusion matrix of proposed methodology.

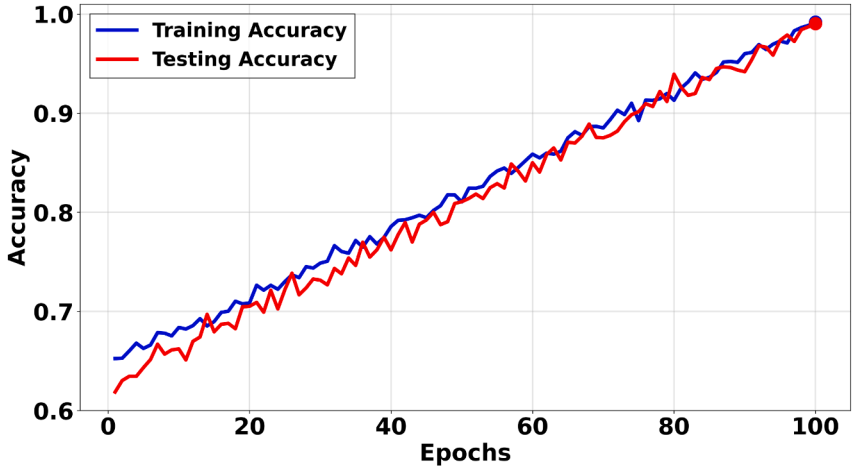


Fig. 11. Accuracy curve vs epochs of proposed methodology.

is the number of buses, 3 is the number of features per bus, and 10 is the number of sliding window samples per scenario, as summarized in Table 3. The confusion matrices (presented in Fig. 10) for the training (80%) and testing (20%) datasets exhibit strong diagonal dominance, indicating high classification accuracy across all major categories, including Healthy Events (HE), Fault Events (FE), and Non-Fault Events (NFE).The NFE class further comprises multiple subcategories, namely DG Penetration (DGP), Loading Change (LC), Varying Power Factor (VPF), and Unbalanced Condition (UC), as detailed in Table 3. Since the NFE class is systematically divided into these balanced subcategories during dataset generation, no significant class imbalance is observed in the confusion matrices. During training, only limited misclassification is observed among the FE, HE, DGP, LC, VPF and UC classes (Fig. 11).

The proposed GNN model demonstrates consistent learning progression, with training and testing accuracy curves closely tracking each other across 100 epochs, indicating no overfitting. At epoch 100, the model achieves a training accuracy of 99.203% (80% data) and testing accuracy of 99.037% (20% data), confirming excellent generalization capability for various classification in the IEEE 33-bus distribution system.

Table 4
Overall performance metrics of the applied GNN model derived from confusion matrices.

Metric	Training (80%)	Testing (20%)
Overall Accuracy	0.99203	0.99037
Micro Precision (Precision$_\mu$)	0.99203	0.99037
Micro Recall (Recall$_\mu$)	0.99203	0.99037
Micro F1-score (F1$_\mu$)	0.99203	0.99037
Weighted Precision (Precision$_w$)	0.99204	0.99039
Weighted Recall (Recall$_w$)	0.99203	0.99037
Weighted F1-score (F1$_w$)	0.99203	0.99038

Table 4 presents the overall classification performance of the proposed GNN model derived from the CM. High accuracies of 0.99203

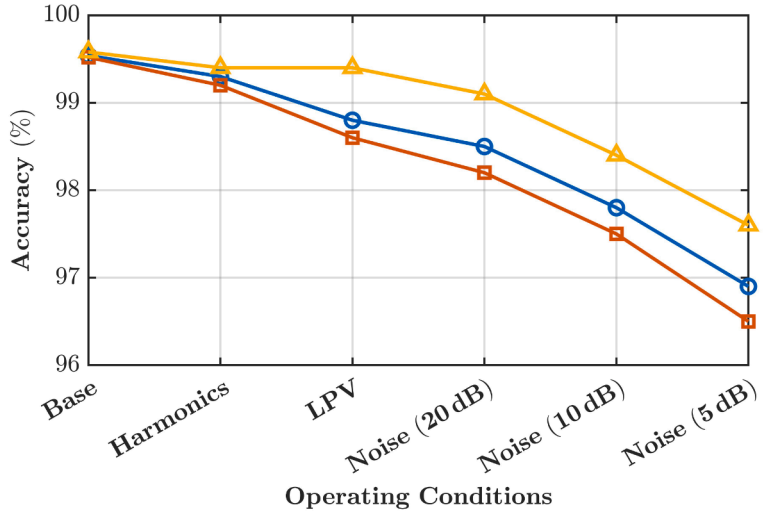


Fig. 12. Accuracy of various scenarios.

Table 5
Classwise performance metrics of the applied GNN model derived from confusion matrices (FE: Fault event, HE: Healthy event, DGP: DGs penetration, LC: Loading change, UPF: Varying power factor, UC: Unbalancing creating).

Metric	Training (80%)			Testing (20%)		
	Precision	Recall	F1-Score	Precision	Recall	F1-Score
FE	0.99481	0.99306	0.99393	0.99394	0.99394	0.99394
HE	0.97395	0.98991	0.98186	0.96517	0.97980	0.97243
DGP	0.99305	0.99221	0.99263	0.99156	0.98906	0.99031
LC	0.99341	0.98990	0.99165	0.99089	0.98889	0.98989
UPF	0.99179	0.99116	0.99147	0.98986	0.98611	0.98798
UC	0.98848	0.99285	0.99066	0.98743	0.99158	0.98950

and 0.99037 are achieved during training and testing, respectively, demonstrating effective learning and strong generalization capability. The equality of the micro-averaged precision, recall, and F_1-score with the overall accuracy indicates uniform predictive performance across all classes, while consistently high weighted-averaged metrics confirm robustness against class imbalance. Table 5 summarizes the overall classwise performance of the proposed model during the training and testing phases, evaluated using Precision, Recall, and F1-score metrics.

4.3. Model sensitivity analysis

4.3.1. Effect of measurement of noise

To examine the effect of measurement noise on the proposed GNN-based FSL framework, additional simulations are carried out by adding additive white Gaussian noise (AWGN) to the three-phase voltage and current signals. This approach emulates practical sensor inaccuracies and communication disturbances encountered in real distribution networks. The sliding window-based features, namely P_n^{SW}, Q_n^{SW}, and PF_n^{SW}, are extracted from the noise-corrupted measurements and supplied as node-level inputs to the trained GNN model for FSL. The localization accuracy obtained at different signal-to-noise ratio (SNR) levels of 5 dB, 10 dB, and 20 dB is presented in Fig. 10.

4.3.2. Impact of line parameter variation

To evaluate the robustness of the proposed methdology, extensive simulation studies are conducted by introducing ±10% variations in the distribution line resistance and inductance. Under these parametric uncertainties, various fault types as well as a NFE are initiated on the line section between Bus 8 and Bus 9, where both the utility grid and DGs units contribute to the fault current. P_n^{SW}, Q_n^{SW}, and PF_n^{SW} are extracted from the measured voltage and current signals and provided as node-level inputs to the trained GNN model for FSL. The classification accuracy under line resistance and inductance variation scenarios is compared with the base case without parameter variations, as shown in Fig. 12. The results indicate that variations in line resistance have a negligible effect on classification accuracy, highlighting the low sensitivity of the proposed GNN model to resistance changes.Overall, the results demonstrate that the sliding-window averaged features P_n^{SW}, Q_n^{SW}, and PF_n^{SW}, when combined with the GNN classifier, ensure reliable and consistent fault classification performance under practical line parameter uncertainties.

Table 6
Comparative analysis of group-wise classification accuracy (%) under different operating conditions.

Operating Condition	FE	NFE	HE
Base Case	99.39	99.15	97.98
Harmonics (Triplen)	99.10	99.00	99.42
Line Parameter Variation	98.80	98.60	99.40
Noise (20 dB)	98.50	98.20	99.10
Noise (10 dB)	97.80	97.50	98.40
Noise (5 dB)	96.90	96.50	97.60

4.3.3. Impact of triplens harmonics

To investigate the influence of triplen harmonic components on the performance of the proposed classification scheme, additional simulation studies are carried out by incorporating triplen harmonics in the voltage and current measurements. These harmonics, which commonly arise due to nonlinear loads and inverter-based distributed generation in distribution networks, introduce waveform distortion and asymmetry in the measured signals. P_n^{SW}, Q_n^{SW}, and PF_n^{SW} features are extracted under both healthy and faulted operating conditions and supplied as node-level inputs to the trained GNN for FSL.

The classification performance under practical operating disturbances, including triplen harmonics, line parameter variation (LPV), and measurement noise, is presented in Fig. 12 and compared with the base case. The results show that the considered disturbances cause only a marginal reduction in classification accuracy, indicating the robustness of the proposed feature set and GNN-based classifier. The use of sliding-window averaged features, namely P_n^{SW}, Q_n^{SW}, and PF_n^{SW}, effectively mitigates the influence of harmonic distortion and measurement noise.Group-wise performance analysis reveals a slight accuracy degradation for the FE and NFE classes, whereas the NE class maintains consistently higher accuracy. These observations confirm the reliability of

Table 7
Summary of ML-based faulty section identification methods (LC: Load change; LPV: Line parameter variation; HRF: High-resistance fault; DGP: DG penetration; TH: Tripen hamonics).

Ref	Algo.	Data	Used	Features	Acc.	Train.	Noise	LC	LPV	HRF	DGP	TH
[25]	SVM + RF	Phasors	V,I	Statistical	96%	Offline	✓	×	×	×	✓	×
[26]	GCN	Phasors	V,I	Node features	98.5%	Offline	✓	×	×	×	✓	×
[27]	GAT	Dynamics	V,I	Graph	98%	Offline	✓	×	×	×	✓	×
[28]	GCN	Phasors	V,I	Local features	95%	Offline	✓	×	×	×	×	×
[29]	CNN	Raw	V,I	Energy matrix	99%	Offline	✓	✓	×	×	×	×
[30]	Sup.	Raw	V	Voltage pattern	95%	Offline	✓	×	×	×	×	×
[31]	CNN	Raw	V	Spectrogram	97%	Off.&RT	✓	×	×	×	×	×
[32]	DNN	Phasors	V,I	CWT scalogram	97%	Offline	✓	✓	✓	✓	✓	×
Proposed	**GNN**	Raw	V,I	$(P^{SW}, Q^{SW}, PF^{SW})$	**99.037%**	Off.&RT	✓	✓	✓	✓	✓	✓

the proposed framework under realistic uncertainties encountered in distribution systems (Table 6).

4.3.4. Comparative analysis of the proposed methodology

The proposed features have shown better performance in locating the faulty line in comparison with the existing smart meter data-based methods, as IBDG penetration causes small changes during a fault. This is already demonstrated in Table 1. Further, in this section, Table 7 provides a comparative evaluation of recent machine learning-based fault section localization (FSL) methods with respect to the employed algorithms, data sources, feature sets, training approaches, and reported localization accuracy. As indicated in the table, the proposed graph-based framework, utilizing sliding-window averaged active power P_n^{SW}, reactive power Q_n^{SW}, and power factor PF_n^{SW}, attains the highest localization accuracy of 99.037%.In addition to accuracy, the performance of the reviewed methods is examined under practical operating variations, including load changes (LC), line parameter variations (LPV), high-resistance faults (HRF), penetration of distributed generation (DGP), and the presence of harmonics and measurement noise. The comparison shows that the proposed method preserves consistent performance across all tested conditions, whereas several existing approaches experience reduced accuracy or operational limitations. Moreover, the proposed framework accommodates both offline and real-time training, which supports its deployment in practical distribution systems. Overall, the results in Table 7 indicate that the proposed approach offers improved accuracy, robustness, and adaptability for fault section localization in realistic distribution network settings.

5. Conclusion and future work

This work presents an advanced ML based FSL framework using a GNN to address the challenges introduced by high renewable penetration, nonlinear network behavior, and measurement uncertainties in modern distribution systems. P_n^{SW}, Q_n^{SW}, and PF_n^{SW} features are extracted from three-phase transient voltage and current measurements acquired from FRTU units and used to train the GNN. The proposed framework reliably distinguishes among FE, NFE (DGP,LC, UPF, UC) and NE operating conditions while accurately localizing faulted sections.

Extensive validation on the IEEE 33-node test system under diverse operating scenarios, including all major fault types, high-resistance faults up to 100 Ω, non-fault disturbances, parameter variations, measurement noise, real-time digital simulation, and real field data, demonstrates high accuracy, precision, recall, and F_1-score. Effective discrimination of NFE cases enables rapid restoration actions, thereby reducing outage duration and associated economic losses. Comparative analysis with other ML methods further confirms the robustness and practical applicability of the proposed approach.

Although the modified IEEE 33-bus test system shown in Fig. 1 operates in a radial configuration under normal conditions, the presence of normally open (NO) switches enables reconfiguration into a meshed network. In the present study, the system is analyzed primarily under radial operation. However, as a part of future work, the proposed feature set will be further evaluated under varying the high impedance fault (HIF), dynamic network reconfiguration scenarios, including transitions from radial to meshed topology. Additionally, the impact of higher penetration levels and diverse types of distributed energy resources (DERs), including wind energy systems, energy storage systems (ESS), and electric vehicle (EV) charging infrastructure, will be incorporated. These studies will provide deeper insight into the adaptability and scalability of the proposed method under evolving active distribution network architectures.

CRediT authorship contribution statement

Archana Tamrakar: Writing – original draft, Visualization, Validation, Software, Methodology; **Saumendra Sarangi:** Writing – review & editing.

Data availability

The data that has been used is confidential.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Archana Tamrakar reports administrative support and equipment (MATLAB SOFTWARE and Typhoon 606 model) supplies were provided by Motilal Nehru National Institute of Technology Allahabad. Archana tamrakar reports a relationship with Motilal Nehru National Institute of Technology Allahabad that includes: employment. Archana Tamrakar has patent none pending to none. I am Archana tamrakar, doing phD in Motilal Nehru National institute of technology, here i am a research scholar in Electrical Engineering department. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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